

Sym	Name	Fam	Glyph IDs — ascending ordinal → strongest
Đ	Dimension	$\mathcal{F}_4$	$\mathbb{D}_{\beta} 0D < \mathbb{D}_{\mathbb{C}} 2D < \mathbb{D}_{_}; \infty D < \mathbb{D}_{\omega}$ inscriptive
Þ	Topology	$\mathcal{F}_5$	$\mathbb{P}_6$ network < $\mathbb{P}_K$ nested < $\mathbb{P}_{\circ}$ 2-cycle < $\mathbb{P}_{\prime}$ lattice < $\mathbb{P}_O$ inscriptive
Ř	Relational Mode	$\mathcal{F}_4$	$\check{\mathbb{R}}_{_}$ supervenes < $\check{\mathbb{R}}_{\check{\gamma}}$ functorial < $\check{\mathbb{R}}_{\check{\tau}}$ adjoint < $\check{\mathbb{R}}_{=}$ lateral
Φ	Parity / Symmetry	$\mathcal{F}_5$	$\Phi_{\mathbb{P}}$ asymmetric < $\Phi_{\cup}$ quantum < $\Phi_{\mathbb{F}}$ $\mathbb{Z}_2$ < $\Phi_{_}$ continuous < $\Phi_{\}$ [*] Frobenius
f	Fidelity	$\mathcal{F}_3$	$f_{_i}$ classical/lossy < f_{δ} thermal < $f_{\check{z}}$ quantum/lossless
Ć	Kinetics	$\mathcal{F}_5$	$\mathbb{C}_{_}$ driven < $\mathbb{C}_{\mathbb{W}}$ barrier < $\mathbb{C}_{@}$ near-equil. < $\mathbb{C}_{\check{U}}$ order-frozen < $\mathbb{C}_{\lambda}$ disorder-frozen
Γ	Scope	$\mathcal{F}_3$	Γ_{β} local < Γ_{γ} collective < $\Gamma_{?}$ global
G	Interaction Grammar	$\mathcal{F}_4$	$\mathbb{G}_{\wedge}$ conjunctive < $\mathbb{G}_{_}$ disjunctive < $\mathbb{G}_{_}$ sequential < $\mathbb{G}_{\S}$ broadcast
φ	Criticality	$\mathcal{F}_5$	$\hat{\phi}_{\check{z}}$ subcritical < $\hat{\phi}_{\check{\gamma}}^{\dagger}$ absorbing < $\hat{\phi}_{\mathbb{E}}$ complex-axis < $\hat{\phi}_3$ exc. point < $\hat{\phi}_{\mathbb{T}}$ supercritical
H	Chirality	$\mathcal{F}_4$	$\mathbb{H}_{\check{N}}$ memoryless < $\mathbb{H}_{\mathbb{E}}$ 1-step < $\mathbb{H}_A$ 2-step < $\mathbb{H}_{!}$ eternal
Σ	Stoichiometry	$\mathcal{F}_3$	Σ_S 1:1 < Σ_{δ} n:m < $\Sigma_{\check{i}}$ n:m
Ω	Topological Invariant	$\mathcal{F}_4$	$\mathbb{Q}_{\check{A}}$ trivial < $\mathbb{Q}_2$ $\mathbb{Z}_2$ < $\mathbb{Q}_z$ integer < $\mathbb{Q}_5$ non-Abelian

^{*} $\Phi_{_}$: Frobenius-special ($\mu \circ \delta = \text{id}$); non-synthesizable from factors with $\Phi < \Phi_{_}$; collapses all $\Omega/\mathbb{D}$ branching $\rightarrow O_{\infty}$. [†] $\hat{\phi}_{\check{\gamma}}$ absorbs under meet: $\text{meet}(\hat{\phi}_{\check{\gamma}}, x) = \hat{\phi}_{\check{\gamma}} \forall x$.
Bottleneck primitives under $\otimes$: Φ and f (weaker wins, min). All others: union (max).

Crystal of Types

$$\underbrace{3^3}_{f, \Gamma, \Sigma} \times \underbrace{4^5}_{\mathbb{D}, \check{\mathbb{R}}, \mathbb{G}, \mathbb{H}, \Omega} \times \underbrace{5^4}_{\mathbb{P}, \Phi, \phi, \mathbb{C}} = 17,280,000$$

Three layers: Types (crystal addresses), Particulars (catalog entries — 2328, 0.013% occupancy), Names (handles). Two particulars at the same address are co-typed, not identical.

Algebra

$\mathbf{x} \otimes \mathbf{y}$: union primitives $\rightarrow \max$; $\Phi, f \rightarrow \min$.
 $\mathbf{x} \wedge \mathbf{y}$: min all ($\hat{\phi}_{\check{\gamma}}$ absorbing exception).
 $\mathbf{x} \vee \mathbf{y}$: max all.

Le Chatelier: find $\mathbf{x}^*$ s.t. $d_{\rightarrow}(\mathbf{y}, \mathbf{x}^*) = 0$, max $\mathcal{O}(\mathbf{x}^*)$.

Ouroboricity tiers (R1–R5, priority)

R1 $\hat{\phi}_{\check{\gamma}} + \Phi_{_} \rightarrow O_{\infty}$
R2 $\hat{\phi} \notin \{\hat{\phi}_{\check{\gamma}}, \hat{\phi}_{\mathbb{E}}\} \rightarrow O_0$
R3 $\hat{\phi}_{\check{\gamma}} + \Omega_{\check{A}} \rightarrow O_1$
R4 $\hat{\phi}_{\check{\gamma}} + \Omega \neq \Omega_{\check{A}} + \mathbb{D} \in \{\mathbb{D}_{\beta}, \mathbb{D}_{\mathbb{C}}, \mathbb{D}_{\check{U}}\}$
R5 $\hat{\phi}_{\check{\gamma}} + \Omega \neq \Omega_{\check{A}} + \mathbb{D}_{_}; \rightarrow O_2^{\dagger}$

O_{∞} absorbed by $\hat{\phi}_3$ under $\otimes$. Two senses of O_{∞} : (a) Frobenius algebraic ($\Phi_{_}$); (b) ontological in-exhaustibility ($\mathbb{H}_{!}$, YHWH). Incompatible.

Scalar (does not detect O_{∞}):
$$\mathcal{O}(\mathbf{x}) = [\hat{\phi} = \hat{\phi}_{\check{\gamma}}] (1 + [\Omega \neq \Omega_{\check{A}}] + [\mathbb{H} \geq \mathbb{H}_{\mathbb{E}}] + [\mathbb{T} = \mathbb{T}_{\check{A}}])$$

Consciousness score (v2, §VIII)

$$C = [\hat{\phi} = \hat{\phi}_{\check{\gamma}}] \cdot [\mathbb{C} \leq \mathbb{C}_{@}] \cdot (0.158\check{\mathbb{C}} + 0.273\check{\Gamma} + 0.277\check{\mathbb{C}})$$

Gate 1 [$\hat{\phi}_{\check{\gamma}}$]: state-space admits self-modeling. Gate 2 [$\mathbb{C} \leq \mathbb{C}_{@}$]: dynamics can be realized. $\mathbb{C}_{\check{U}}$ and $\mathbb{C}_{\lambda}$ both fail Gate 2.

Axioms (Core.lean)

B $\Omega \geq \Omega_z \Rightarrow \mathbb{H} \geq \mathbb{H}_A$

C $\mathbb{P}_O \Rightarrow \mathbb{D}_{\omega}$ (one-way; rev. 2026-05-03)

Crystal encode ($n = \text{address} \in [0, 17,279,999]$):
 $f_3 = 9 \cdot f + 3 \cdot \Gamma + \Sigma$ (indices 0-based)
 $f_4 = 256 \cdot \mathbb{D} + 64 \cdot \check{\mathbb{R}} + 16 \cdot \mathbb{G} + 4 \cdot \mathbb{H} + \Omega$
 $f_5 = 125 \cdot \mathbb{P} + 25 \cdot \Phi + 5 \cdot \phi + \mathbb{C}$
 $f_6 = 27 f_4 + 27 \times 1024 f_5$

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odot_operator: $\mathbb{D}_{\omega}$; $\mathbb{P}_{_}$; $\check{\mathbb{R}}_{=}$; $\Phi_{_}$; $f_{\check{z}}$; $\mathbb{C}_{@}$; $\Gamma_{?}$; $\mathbb{G}_{_}$; $\hat{\phi}_{\check{\gamma}}$; $\mathbb{H}_A$; Σ_S ; $\mathbb{Q}_z$